👩‍🏫 Staff Guide · Biology A · Blood Sugar & Diabetes (Level 3)

Level 3 = resource-room students who can work close to the gen ed lesson. Same topics and order as Ms. Dunham's class; more reading and writing than Level 2.

1. How the pieces line up

Class lesson	Level 3 piece
A1 What is going on (CDC data)	Reading part 5; notes section C
A3/A6 Homeostasis (blood sugar game)	Reading parts 1-2; notes section A
ClEvR quizzes #1-3	Reading parts 3-4; notes section B (graph math)
A7 Genetic & environmental risks; A8 Solutions	Notes section C
A10 Review + Diabetes Test	Quick check + unit game

2. Supports built in

- Reading and notes each have a web copy with a 🔊 Read aloud button (highlights as it reads)
- Key-word box at the top of the notes; word problems have a worked example first
- Less filled in than Level 2: short answers, tables, a drawing
- No slide decks: students follow Ms. Dunham's slides with the Level 3 notes

3. Files

- Reading (+ 🔊 web copy)
- Guided notes + key (+ 🔊 web copy)
- Quick check + key
- Unit game: 10 questions

4. What to grade

- Guided notes stand in for the matching class notes or worksheet (same points when complete)
- Quick check: score it, then scale to the class quiz points
- Reading check questions: completion points
- Modified grading only as the IEP allows. Check with the case manager.

5. Who to ask

- Mr. Driver (SLE): changes and IEP questions.
- Class content: Ms. Dunham.

💗 Sensitive topic note

Some students or their families may live with diabetes. Keep examples general, never ask a student to share health information, and treat type 1 as not caused by food choices.

Blood Sugar & Diabetes: Guided Notes

Biology A · Unit 6 · Level 3 (closer to the class work) · goes with Unit 3 Diabetes: A1 CDC data, A3/A6 Homeostasis, ClEvR quizzes, A7 Risks, A8 Solutions, A10 Review, Diabetes Test

Name: _____

Use the class slides and your reading. Key words:

homeostasis | *blood glucose* | *insulin* | *glucagon* | *pancreas* | *feedback loop* | *type 1* | *type 2* | *risk factor*

A. Homeostasis (A3/A6)

1. Keeping inside conditions balanced is _____.

2. Fill in the table.

Blood sugar is…	Hormone released	What happens
HIGH (after a meal)		
LOW (skipped a meal)		

B. Diabetes (ClEvR, A1)

3. Most cases (90-95%) are type _____.

4. **Reading the ClEvR #2 graph**

> Example: 210 − 130 = 80 mg/dL higher

a) Diabetic: 240. Non-diabetic: 135. Difference = _____

b) Which number is above the 200 line? _____

5. Why do sugary drinks spike blood sugar more than beans?

_____

_____

C. Risks and solutions (A7, A8)

6. Fill in the table.

Risk factor	Genetic or environmental?	Can it change?
family history		
daily activity		
food in local stores		

7. Describe one community solution and how it lowers risk.

_____

_____

Blood Sugar & Diabetes: Reading

Biology A · Unit 6 · Level 3 (closer to the class work) · goes with Unit 3 Diabetes: A1 CDC data, A3/A6 Homeostasis, ClEvR quizzes, A7 Risks, A8 Solutions, A10 Review, Diabetes Test

Name: _____

Keeping Blood Sugar in Balance

Homeostasis is the body keeping its inside conditions balanced, the way a thermostat keeps a room at the right temperature. Blood sugar (blood glucose) goes up after we eat. A feedback loop keeps it in a healthy range.

When blood sugar is high, the pancreas releases insulin. Insulin works like a key: it unlocks cells so glucose can go in, and blood sugar goes down. When blood sugar is low, the pancreas releases glucagon, which lets stored sugar out into the blood, and blood sugar goes up.

Some foods raise blood sugar fast. Sugary drinks and white bread make it spike. Beans, vegetables and whole grains raise it slowly because of fiber. Sugar in whole fruit comes with fiber too.

Diabetes is a disease where blood sugar stays too high for too long. In type 1 diabetes, the pancreas makes little or no insulin; it often starts young and is not caused by food choices. In type 2 diabetes (90-95% of cases), cells stop answering insulin well. In ClEvR #2, diabetic blood sugar three hours after eating was 200 or more, compared with about 120-140 without diabetes.

Risk factors can be genetic (family history, which we cannot change) or environmental (food, activity, sleep, money and where people live, some of which can change). CDC data shows higher rates where people have less money or schooling. Solutions can help people and whole communities: drinking water instead of sugary drinks, more chances to move, clear food labels and medical treatment.

Check your reading

1. What does insulin do?

2. How is type 1 different from type 2 diabetes?

3. Give one genetic and one environmental risk factor.

Blood Sugar & Diabetes: Quick Check

Biology A · Unit 6 · Level 3 (closer to the class work) · goes with Unit 3 Diabetes: A1 CDC data, A3/A6 Homeostasis, ClEvR quizzes, A7 Risks, A8 Solutions, A10 Review, Diabetes Test

Name: _____

Multiple choice and short answer. You may have it read aloud.

1. Insulin is made by the…

A. Pancreas | B. Liver | C. Lungs

2. When blood sugar is low, the body releases…

A. Insulin | B. Saliva | C. Glucagon

3. Type 1 diabetes is…

A. Always caused by sugar | B. The most common type | C. Not caused by food choices

4. Which raises blood sugar slowly?

A. Soda | B. Beans | C. White bread

5. Explain homeostasis using a thermostat example.

6. Name one environmental risk factor for type 2 diabetes that a community could change.

🩸 Blood Sugar & Diabetes: Guided Notes · ANSWER KEY

Biology A · Unit 6 · Level 3 (closer to the class work) · goes with Unit 3 Diabetes: A1 CDC data, A3/A6 Homeostasis, ClEvR quizzes, A7 Risks, A8 Solutions, A10 Review, Diabetes Test

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homeostasis | *blood glucose* | *insulin* | *glucagon* | *pancreas* | *feedback loop* | *type 1* | *type 2* | *risk factor*

A. Homeostasis (A3/A6)

1. Keeping inside conditions balanced is **homeostasis** .

2. Fill in the table.

Blood sugar is…	Hormone released	What happens
HIGH (after a meal)	**insulin**	**glucose goes into cells; blood sugar goes down**
LOW (skipped a meal)	**glucagon**	**stored sugar is released; blood sugar goes up**

B. Diabetes (ClEvR, A1)

3. Most cases (90-95%) are type **2** .

4. **Reading the ClEvR #2 graph**

> Example: 210 − 130 = 80 mg/dL higher

a) Diabetic: 240. Non-diabetic: 135. Difference = **105 mg/dL**

b) Which number is above the 200 line? **240 (diabetic)**

5. Why do sugary drinks spike blood sugar more than beans?

Sugary drinks have sugar with no fiber, so it enters the blood fast; beans have fiber and raise it slowly.

C. Risks and solutions (A7, A8)

6. Fill in the table.

Risk factor	Genetic or environmental?	Can it change?
family history	**genetic**	**no**
daily activity	**environmental**	**yes**
food in local stores	**environmental**	**yes, with community help**

7. Describe one community solution and how it lowers risk.

Example: water fountains at school so students drink water instead of soda, lowering sugar spikes.

🔑 *Key written by our team from Ms. Dunham's materials, please check.*

Blood Sugar & Diabetes: Reading · ANSWER KEY

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Check your reading

1. What does insulin do?
It unlocks cells so glucose can go in, which lowers blood sugar.

2. How is type 1 different from type 2 diabetes?
Type 1: the pancreas makes little or no insulin. Type 2: cells stop answering insulin well.

3. Give one genetic and one environmental risk factor.
Genetic: family history. Environmental: food, activity, sleep, money, place.

🩸 Blood Sugar & Diabetes: Quick Check · ANSWER KEY

Biology A · Unit 6 · Level 3 (closer to the class work) · goes with Unit 3 Diabetes: A1 CDC data, A3/A6 Homeostasis, ClEvR quizzes, A7 Risks, A8 Solutions, A10 Review, Diabetes Test

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Multiple choice and short answer. You may have it read aloud.

1. Insulin is made by the…

A. Pancreas | B. Liver | C. Lungs

2. When blood sugar is low, the body releases…

A. Insulin | B. Saliva | **C. Glucagon**

3. Type 1 diabetes is…

A. Always caused by sugar | B. The most common type | **C. Not caused by food choices**

4. Which raises blood sugar slowly?

A. Soda | **B. Beans** | C. White bread

5. Explain homeostasis using a thermostat example.

A thermostat heats when cold and cools when hot to stay balanced; the body raises or lowers blood sugar the same way.

6. Name one environmental risk factor for type 2 diabetes that a community could change.

Any: access to healthy food, places to exercise, water instead of soda, health care.

🔑 *Key written by our team from Ms. Dunham's quizzes and notes, please check.*